

Out of all the Disney movies of 2008, *Cars* was (and still is) my favorite of all time. The high octane racing scenes get your blood pressure up as you watch on the edge of your seat, waiting to see if McQueen will win the Piston cup before Chick Hicks and The King catch up to him. When I took on my FSAE onboarding project — reading raw parquet files and performing data manipulation using Python—that lifelong passion turned into hands-on engineering experience. Over the course of the project, I dove headfirst into Git, learned how to handle Parquet files, and discovered just how tricky and rewarding it is to code in Python.

Despite having previous coding experience, my progress on this onboarding project was slower than I anticipated or expected. I was constantly held back by how much I didn't know, frequently turning to YouTube to figure out how to do a function. At the beginning of the project, before I ever touched a terminal, I started with a YouTube video on the basics of coding in Python. I was surprised; while it was similar to Java in a few ways, it was also very different. To start, coding in Python is very colorful, which makes it a lot easier to read the text (although you are still reading big blocks of text). Another thing that was a bit of a slow adjustment for me was the fact that python does not require a semicolon at the end of a line, and instead of curly brackets, they use indentation which also improves the readability. Another part of this experience was learning GitHub and Git functions. Initially, it was frustrating and confusing when I couldn't do what I wanted. After a few Youtube videos, however, I started getting familiar with basic Git functions, and after using Git and playing around with it, I became familiar with basic functions, what they do, and how to execute them (although I still have trouble from time to time). I find the parquet files themselves very interesting, since all of the data is very neatly arranged in an array. One nice thing about polars is that there are functions that allow you to draw out and isolate a single data frame from the main data frame, which I did in order to convert the raw speed data from RPM to MPH.

While I might not be tuning engines for the Piston Cup, wrestling with Python syntax, battling Git repositories, and manipulating parquet files taught me more than any classroom ever could. I'm walking away from this project with an excitement and drive to learn more about this and further my knowledge, whether that be in more complex Python functions, furthering my knowledge of Git, or executing full on coding projects and storing them on Git. If selected, I am ready to bring my high octane, high energy vibe into the team to push FS-6 to new heights.